

Logistic Regression Analysis

Comprehensive Guide with Practical Applications

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SUST Statistics

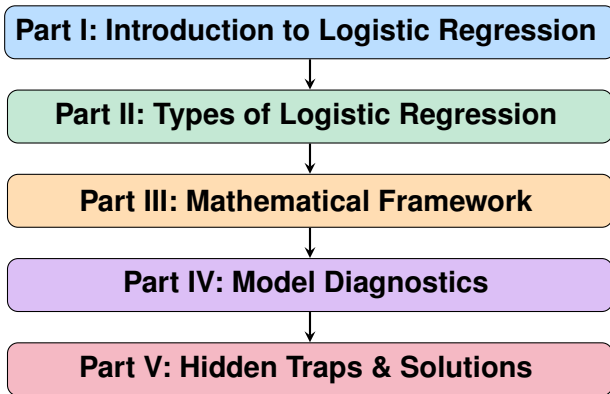


ORCID

THE ARCHITECTURE SERIES

Production-grade rigour starts where the textbooks stop.

Today's Roadmap

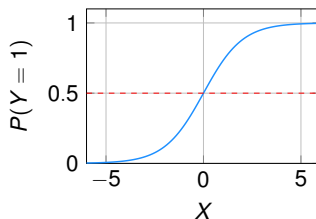


What is Logistic Regression?

Definition

Logistic regression is a **statistical method** for:

- **Binary classification** problems
- Predicting **probability** of an event
- Understanding **relationships** between variables



The Logit Transformation

Formula

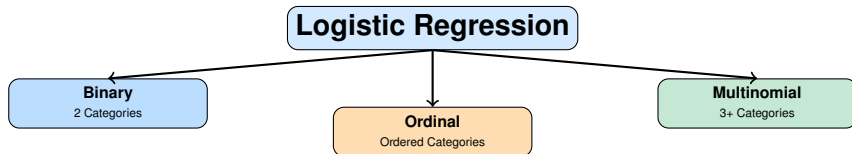
$$\text{logit}(p) = \ln \left(\frac{p}{1-p} \right) \quad (1)$$

- p = Probability of event
- $\frac{p}{1-p}$ = Odds

Example Values

Prob	Odds	Log-odds
0.1	0.11	-2.20
0.3	0.43	-0.85
0.5	1.00	0.00
0.7	2.33	0.85
0.9	9.00	2.20

Types of Logistic Regression



Binary Logistic Regression

Model

$$P(Y = 1|X) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X)}} \quad (2)$$

Logit Form:

$$\log\left(\frac{p}{1-p}\right) = \beta_0 + \beta_1 X \quad (3)$$

Interpretation

- β_0 : Intercept
- β_1 : Log-odds change
- e^{β_1} : **Odds Ratio**
- $OR > 1$: Positive association
- $OR < 1$: Negative association

Odds Ratio Interpretation

What is Odds Ratio?

$$OR = \frac{\text{Odds in group 1}}{\text{Odds in group 2}} = \frac{p_1/(1 - p_1)}{p_2/(1 - p_2)} \quad (4)$$

OR Value	Interpretation	Effect
OR = 1	No association	Null
OR > 1	Increased odds	Risk factor
OR < 1	Decreased odds	Protective factor

Example

If OR = 2.5 for "Urban vs Rural":
Urban children have 2.5 times higher odds of ECE

Multinomial Logistic Regression

Model

For J categories:

$$P(Y = j|X) = \frac{e^{\beta_j X}}{1 + \sum_{l=1}^{J-1} e^{\beta_l X}} \quad (5)$$

For reference:

$$P(Y = J|X) = \frac{1}{1 + \sum_{l=1}^{J-1} e^{\beta_l X}} \quad (6)$$

Interpretation

- Choose **reference** category
- $e^{\beta_{jk}}$: Relative Risk Ratio
- $RRR > 1$: More likely
- $RRR < 1$: Less likely

Ordinal Logistic Regression

Proportional Odds Model

$$P(Y \leq j|X) = \frac{1}{1 + e^{-(\alpha_j - \beta X)}} \quad (7)$$

Where $j = 1, 2, \dots, J - 1$

Key Points

- α_j : Thresholds
- β : Common effect
- **Assumption**: Effect is constant
- Check with Brant test

Maximum Likelihood Estimation (MLE)

Likelihood

$$\mathcal{L}(\beta) = \prod_{i=1}^n P_i^{y_i} (1 - P_i)^{1-y_i} \quad (8)$$

Log-Likelihood

$$\ell(\beta) = \sum_{i=1}^n [y_i \log(P_i) + (1 - y_i) \log(1 - P_i)] \quad (9)$$

How It Works

- 1 Start with initial β values
- 2 Calculate likelihood
- 3 Use iterative algorithm
- 4 Find β that **maximizes** likelihood

TIER 1: The Standard Checks

Linearity

Check linearity in log-odds for continuous predictors.

- If line bends: Use polynomial/spline

Independence

This is a **design issue**:

- Use mixed-effects models
- Clustered standard errors

Multicollinearity

Check **VIF**:

- $VIF > 5$: Warning
- $VIF > 10$: Red flag

Influential Outliers

Calculate **Cook's Distance**:

- $D > 1$: Concern

Multicollinearity - VIF

VIF Formula

$$VIF_j = \frac{1}{1 - R_j^2} \quad (10)$$

Interpretation

VIF	Status	Action
1-5	Acceptable	No action
5-10	Warning	Consider combining
>10	Critical	Remove variable

```
from statsmodels.stats.outliers_influence
import variance_inflation_factor

vif = pd.DataFrame()
vif['Variable'] = X.columns
vif['VIF'] = [variance_inflation_factor(
    X.values, i)
    for i in range(X.shape[1])]
```

Python Code

Influential Outliers - Cook's Distance

Cook's Distance

$$D_i = \frac{\sum_{j=1}^n (\hat{y}_j - \hat{y}_{j(i)})^2}{p \cdot \text{MSE}} \quad (11)$$

```
model <- glm(y ~ x1 + x2,  
             family = binomial)  
cooks <- cooks.distance(model)  
influential <- which(cooks > 1)  
plot(cooks, type="h")  
abline(h=1, col="red")
```

R Code

Interpretation

D Value	Interpretation
D < 0.5	Acceptable
0.5-1	Moderate concern
D > 1	Critical

TIER 2: The Hidden Traps

Perfect Separation

Signs:

- OR in hundreds
- Huge confidence intervals

Solution: Firth's penalized likelihood

"One in the Box"

Situation: Categories with low counts

- No mix of outcomes
- Unstable estimates

Solution: Collapse categories

Too Few Events (EPV)

Aim: At least 10 events per variable

- $EPV < 10$: Warning

Perfect Separation

Example

Group	Event	No Event	Total
A	20	0	20
B	5	25	30

Problem

- $OR = \infty$
- Model fails to converge

Solution: Firth's Method

$$\beta_{\text{Firth}} = \arg \max_{\beta} [\ell(\beta) + 0.5 \log |I(\beta)|] \quad (12)$$

```
library(logistf)
model <- logistf(y ~ x1 + x2)
```

R Code

Events Per Variable (EPV)

What is EPV?

$$\text{EPV} = \frac{\text{Number of events}}{\text{Number of predictors}}$$

Guidelines

EPV	Recommendation
-----	----------------

< 5	Too low
5-10	Marginal
≥ 10	Adequate

Why It Matters

Low EPV leads to:

- Overfitting
- Unstable coefficients
- Wide confidence intervals

Solutions

- 1 Simplify model
- 2 Use penalized regression
- 3 Collect more data

Goodness of Fit Tests

Hosmer-Lemeshow Test

$$\hat{C} = \sum_{g=1}^G \frac{(O_g - E_g)^2}{E_g(1 - E_g/n_g)} \quad (13)$$

- H_0 : Model fits well
- $p > 0.05$: Good fit

Limitations

- Sensitive to binning
- Sensitive to sample size

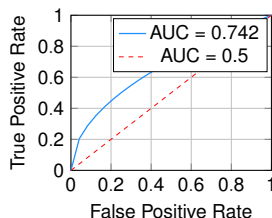
Alternatives

- Likelihood Ratio Test
- AIC / BIC
- Calibration plots

Model Discrimination - AUC-ROC

AUC Definition

$$\text{AUC} = \int_0^1 \text{TPR}(\text{FPR}) d(\text{FPR}) \quad (14)$$



Interpretation

AUC	Interpretation
0.5	No discrimination
0.7-0.8	Acceptable
0.8-0.9	Excellent
>0.9	Outstanding

Example 1: Binary Logistic Regression - ECE

Context

What factors predict Early Childhood Education (ECE) participation in Bangladesh?

Model

$$\begin{aligned} \log \left(\frac{P(\text{ECE} = 1)}{1 - P(\text{ECE} = 1)} \right) = & \beta_0 + \beta_1(\text{WASH}) + \beta_2(\text{Wealth}) \\ & + \beta_3(\text{Maternal Education}) + \beta_4(\text{Urban}) \\ & + \beta_5(\text{Media Access}) + \beta_6(\text{Child Age}) \end{aligned} \quad (15)$$

Table: Multivariable Logistic Regression Results

Predictor	AOR	95% CI	p-value
WASH Index	1.45	(1.21, 1.74)	<0.001
Wealth (Middle)	1.82	(1.45, 2.28)	<0.001
Wealth (Highest)	2.94	(2.31, 3.74)	<0.001
Maternal Education	1.12	(1.08, 1.16)	<0.001
Urban Residence	1.38	(1.12, 1.70)	0.002
Media Access	1.25	(1.04, 1.50)	0.018
Child Age	1.03	(1.02, 1.04)	<0.001

Model: H-L $\chi^2 = 8.42$, df=8, p=0.393

Discrimination: AUC = 0.742

Example 2: Multinomial Logistic Regression

Context

What factors predict depression severity levels? (None, Mild, Moderate, Severe)

Model

$$\log \left(\frac{P(Y = j)}{P(Y = \text{None})} \right) = \beta_{0j} + \beta_{1j}X_1 + \dots \quad (16)$$

for $j = \{\text{Mild, Moderate, Severe}\}$

Multinomial Results

Table: Multinomial Regression for Depression Severity

Predictor	Mild vs None	Moderate vs None	Severe vs None
Age	1.02 (0.98,1.06)	1.04* (1.00,1.08)	1.06** (1.02,1.10)
Female	1.32** (1.12,1.56)	1.58*** (1.28,1.95)	1.89*** (1.45,2.46)
Low Income	1.45*** (1.22,1.72)	1.82*** (1.47,2.25)	2.34*** (1.78,3.08)
Urban	0.88 (0.74,1.05)	0.82* (0.67,1.00)	0.74** (0.58,0.94)
Decision Autonomy	0.76*** (0.66,0.88)	0.65*** (0.54,0.78)	0.52*** (0.41,0.66)

Model: $\chi^2 = 284.5$, df=15, $p < 0.001$

Pseudo R^2 : McFadden = 0.124

TIER 3: The Architect's Mindset

Diagnose the Dominant Trap

Which issue is most likely to break your model in production?

- Perfect separation?
- Too few events?
- Multicollinearity?

Remember the Business Cost

A model can pass all statistical tests and still be:

- **Operationally wrong**
- **Unusable in practice**

Check Model Fit

Don't just look at coefficients!

- Likelihood Ratio Test
- Hosmer-Lemeshow Test

Key Questions

- 1 Does it solve the problem?
- 2 Can it be implemented?
- 3 What are the costs of errors?

Summary: Key Diagnostic Tests

Table: Key Diagnostic Tests

Test	Purpose	Threshold
VIF	Multicollinearity	>5 Warning, >10 Critical
Cook's Distance	Influential outliers	$D > 1$
Hosmer-Lemeshow	Calibration	$p > 0.05$
AUC-ROC	Discrimination	0.7-0.8 Acceptable
Likelihood Ratio	Overall fit	$p < 0.05$
EPV	Sample adequacy	> 10

Implementation: Python

```
import pandas as pd
import numpy as np
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import roc_auc_score
import statsmodels.api as sm

# Prepare data
X = df[['maternal_edu', 'wealth', 'urban']]
y = df['ece_participation']

# Statsmodels
X_const = sm.add_constant(X)
model = sm.Logit(y, X_const).fit()

# Results
print(model.summary())

# Odds Ratios
or_vals = np.exp(model.params)
print(or_vals)

# AUC
y_pred = model.predict(X_const)
auc = roc_auc_score(y, y_pred)
print(f"AUC: {auc:.4f}")
```

Binary Logistic Regression

Implementation: R

```
library(tidyverse)
library(car)
library(pROC)
library(ResourceSelection)

# Fit model
model <- glm(ece ~ maternal_edu +
             wealth + urban,
             data = df,
             family = binomial())

# Summary
summary(model)

# Odds Ratios
or_ci <- exp(cbind(OR = coef(model),
                  confint(model)))

# VIF
vif(model)

# Hosmer-Lemeshow
hl_test <- hoslem.test(df$ece,
                      fitted(model), g=10)

# AUC
roc_obj <- roc(df$ece, fitted(model))
auc(roc_obj)
```

Binary Logistic Regression

Common Errors & Solutions

Error: Model fails to converge **Solution:** Check for perfect separation

Error: Huge confidence intervals **Solution:** Check EPV or add data

Error: "NA" values in coefficients **Solution:** Check multicollinearity

Error: Poor fit ($H-L\ p < 0.05$) **Solution:** Add interactions or nonlinear terms

Resources

Recommended Books

- **Applied Logistic Regression**
Hosmer, Lemeshow & Sturdivant
- **Categorical Data Analysis**
Agresti
- **Regression Modeling Strategies**
Harrell

Online Resources

-  R Project
-  Scikit-learn
-  GitHub
-  Stack Overflow

Production-grade rigour starts where the textbooks stop.

Questions?

"A model can pass all statistical tests and still be operationally wrong."

Thank You!

Happy Modeling!

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Turning data into decisions that save lives